OpenHamClock

A free, open-source dashboard in your web browser that puts what's happening on the air on one screen: space weather, live spots, propagation for your location, parks and summits, DXpeditions and contests. **openhamclock.com** · no account, no download, no ads.

Getting started

1. Open **openhamclock.com** in any browser: computer, tablet or phone.
2. Enter your **callsign** and **grid square**. "Use my current location" fills in the grid for you.
3. You're on the **Modern** layout (the default). Others: Settings → Display → Layout.
4. **Click your callsign** (top left) to open Settings at any time.

Settings are saved in your browser, on that device. To install it like an app: iPhone Safari Share → Add to Home Screen; Android Chrome ⋮ → Install app; Chrome/Edge on a computer, the install icon in the address bar.

> **RULE OF THUMB** High **SFI** and low **K** = a good HF day.



The Modern layout

1. **Header**: clocks and space weather
2. **DE / DX**: your station and your target
3. **Solar**: the sun, its numbers, flares
4. **Propagation**: which bands work, and when
5. **Map**
6. **DX cluster**: who's being spotted
7. **PSKReporter**: who hears you
8. **DXpeditions, parks & summits, contests**

Left side = conditions. Right side = activity. Middle = where it all is.

NOTES

1 Header: clocks and space weather

UTC is the time used by every log, spot, contest and DXpedition (EDT = UTC−4; EST = UTC−5). Click **LOCAL** to switch 12/24 hour. Click your **callsign** for Settings, the **version** for what's new.

	WHAT IT IS	HOW TO READ IT (APP COLORS)
SFI	Solar Flux Index: the sun's radio output at 10.7 cm; a stand-in for how hard it's ionizing the upper atmosphere	Higher is better, especially 15–10 m. ~70 quiet sun · 120+ good · 150+ excellent
K	Planetary K-index: geomagnetic disturbance, 0–9, over 3 hours	Low is better. 0–2 quiet. 4+ shows red: absorption, noise, aurora
SSN	Sunspot number	Rises and falls with SFI over the 11-year cycle
A	A-index: the whole day's geomagnetic disturbance in one number	<10 green · 10–19 amber · 20+ red
SW	Solar wind speed, km/s	<500 green · 500–699 amber · 700+ red. Fast wind stirs up the field
Word	Geomagnetic field summary from N0NBH (HamQSL)	QUIET green · ACTIVE or any STORM red · anything else amber

2 DE: your station

- "DE" is telegraph shorthand for "from": **your end**.
- **Grid square**, e.g. FN20nc: FN = 20°×10° field, 20 = 2°×1° square, nc = a few miles. Most of Lower Bucks is FN20.
- Lat/long: every distance and heading is measured from here.
- **Sunrise → sunset** (local). Low bands improve after dark; gray-line openings happen around these times.

2 DX: your target

- **Local time** at their end
- **Beam Dir**: short-path great-circle heading. Japan from here is ~332°, over the pole.
- **Distance** along that path, and their sunrise/sunset in UTC
- **Set it** by clicking the map, typing a grid or callsign + Enter, picking from **DXCC**, or clicking any spot. ★ = favorites, 🔒 = stop map clicks moving it.

5 The map

- Shaded side = night. Its edge is the **gray line**: low-band signals travel unusually far along it.
- ☀ = where the sun is directly overhead; the moon icon = where the moon is.
- **Each line is a spot**, from the station that heard to the station heard, colored by band (same colors everywhere in the app).
- The **band bar** at the bottom filters the map to one band. The ticker is DX news.
- **Flat**: everyday view; choose a basemap.
- **Azimuthal**: centered on you; a straight line out from the center is your true beam heading.
- **3D**: a globe.
- Left toolbar: zoom, lock, show/hide callsigns, night darkness.

NOTES

3 Solar pane: four views (button at top right; ▶ rotates them)

- **Image**: live from NASA's Solar Dynamics Observatory. Dark patches are **coronal holes**, the source of fast solar wind that arrives 2–4 days later (watch K climb).
- **Indices**: SFI, K, SSN, solar wind with a word rating, a trend, and a 24-hour K forecast.
- **Lunar**: moon phase and position, for EME.

X-ray: flares. Classes **B → C → M → X**, each 10× the one before.

- B, C: background, little effect.
- **M, X**: HF **radio blackout** on the sunlit side of Earth, starting within minutes and lasting minutes to hours.

Bands die at midday? Check X-ray before you check the coax.

4 Propagation pane: which bands, to there, and when

Chart view: a prediction for the path **from your DE to your DX**. Rows = bands, columns = 24 hours (UTC); the white box is now. Green = likely, yellow/orange = possible, red/dark = unlikely. It uses the ITU's HF model (ITU-R P.533, the physics behind VOACAP), calculated in your browser with your mode, power and antenna (default SSB, 100 W).

Example (FN20 → Tokyo, SSB, Oct 19): 40/30 m best ~08–14 UTC (our night); 17/15 m window ~21–00 UTC (5–8 PM EDT).

MUF and LUF

- **MUF**, maximum usable frequency: the highest frequency the ionosphere typically bends back on this path now. Well above it, the signal goes into space.
- **LUF**, lowest usable frequency: below it, daytime absorption eats the signal.
- Bands between them *can* work; signal strength decides whether they do.

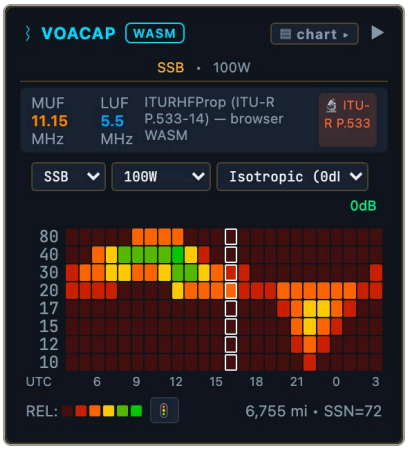


Chart view: FN20 → Tokyo, SSB, 100 W

MODE MATTERS FT8 decodes far below the noise; SSB needs a signal well above it. Switch the mode to FT8 and "closed" bands often open.

GENERAL VS. YOUR PATH "15 m GOOD" means open somewhere. The chart says whether *your* signal reaches *that* place. Use the first to decide whether to get on, the second to decide where to point.

Bars view: the same prediction for this hour, best band first.

Bands view: N0NBH's general GOOD / FAIR / POOR per band, day and night, plus VHF (aurora, sporadic-E) and noise.

NOTES

6 DX cluster

A **spot** is a report that a station was heard on a frequency, from other hams or from automatic receivers (the **Reverse Beacon Network**, RBN), shared over the worldwide DX cluster network.

Row: frequency (colored by band) · callsign · mode · *de* spotter · age.

- **Click a row**: makes it your DX target and shows it on the map.
- 🎧 : listen on a nearby web receiver (KiwiSDR), already tuned.
- **50/162**: rows shown / spots received. Search and filters narrow it.
- With your log imported: `NEW` new country · `BAND` new on this band · `DUPE` already worked · `✓` worked on another band/mode

7 PSKReporter

Digital-mode software (WSJT-X, JTDX…) reports every station it decodes to PSKReporter.info.

- ▲ **Heard**: stations that decoded *you*. ▼ **Hearing**: stations you decoded.
- Numbers are SNR in dB. Gold ★ = you hear each other: a contact is likely.
- "Waiting for spots… (TX to see reports)" = you haven't transmitted yet. Call CQ on FT8 for a few minutes and watch the map: a free antenna test.

8 Programs and calendars

- **POTA** Parks on the Air · **SOTA** Summits on the Air
- **WWFF** World Wide Flora & Fauna · **WWBOTA** Bunkers on the Air
- Row: activator · reference (e.g. US-11290 is a park; hover for its name) · frequency · UTC time · comment ("RBN" = heard by a skimmer)
- **DXpeditions** (NG3K calendar): ● NOW on the air today; SOON coming up.
- **Contests** (WA7BNM calendar): LIVE with time left; upcoming with start time.

Ten seconds before you turn the radio on

1. **Header**: SFI up, K low?
2. **X-ray**: any M or X flare? Day side may be blacked out.
3. **Propagation**: set a DX target; read the chart for band and time.
4. **Map + cluster**: is anyone actually getting through? Filter to that band.
5. **Listen**: 🎧 on a spot, then tune the radio. POTA = easiest contacts.

Settings (icons down the left edge)

Station call, grid, units · **Display** layout, theme, text size · **Map Layers** MUF map, aurora, radar, grid squares · **Satellites** · **Alerts** sounds/notifications · **Profiles** save & share setups · **Integrations** logbooks, QRZ · **Help** the full manual

Ctrl+K (⌘K on Mac): type what you want, e.g. "aurora", "classic".
Other layouts: Classic, Dockable (80+ panes), Contest, EmComm and more.

NOTES